

MATH GU4053 INTRO TO ALGEBRAIC TOPOLOGY

IN-CLASS MIDTERM, 2:40-3:55 ON 2/23/26

6 numbered problems

Closed-book; no notes, computers, calculators, nor visible phones

Unless explicitly stated otherwise, you may use any result proved in class. If your solution makes use of a theorem from class, make sure to explain why it applies (e.g., if the theorem has hypotheses).

Solutions based on an advanced result which has not been covered in this class, or one of its prerequisites, will not receive full credit. (Especially if the use of an advanced result significantly changes the difficulty of the problem.)

It is not sufficient to state that a problem is identical to one from a previous homework or practice exam, when that is the case.

If you run out of space, use the back side of the paper and indicate this clearly.

Name: _____

Uni: _____

1. (a) State the definition of what it means for two spaces X and Y to be homotopy equivalent.

(b) State the definition of what it means for a space to be contractible.

- (c) Let $X = S^2 \setminus \{N, S\}$, where $S^2 = \{v \in \mathbb{R}^3 \mid |v| = 1\}$ and $N = (0, 0, 1) \in S^2$ and $S = (0, 0, -1)$. Let $Y = \mathbb{R}^3 \setminus \{(x, y, z) \in \mathbb{R}^3 \mid x = 0 = y\}$. Prove that X and Y are homotopy equivalent.

2. (a) State the definition of what it means for $p : X \rightarrow Y$ to be a covering map.

- (b) True or false: if $p : X \rightarrow Y$ is a covering and $y \in Y$, and $F = p^{-1}(\{y\}) \subset X$, then the subspace topology on $F \subset X$ agrees with the discrete topology.
Justify your answer.

3. Let G be a group, X a topological space, and $G \times X \rightarrow X$ a continuous action.

(a) State the definition of what it means for the action to be a covering space action.

(b) Let $G \times X \rightarrow X$ be a covering space action and let $Y = X/G$ denote the quotient space. In class we constructed an isomorphism between G and the fundamental group of Y under suitable assumptions—what were the assumptions?

You do not need to justify your answer to this question.

(c) Under the assumptions where we proved that G is isomorphic to the fundamental group of Y , explain what the isomorphism is.

You should explain how the isomorphism is defined. You do not need to prove that it is well defined, nor that it is an isomorphism

4. (a) Define an action of $G = \mathbb{Z}$ on $X = \mathbb{R}^2 \setminus \{0\}$ by $n.x = 2^n x$ for $x \in X$ and $n \in G$. It may be used without proof that this is a continuous action.

Prove that this is a covering space action.

- (b) Let $v_0 \in \mathbb{R}^n$, let $X = \mathbb{R}^n$, and let $G = \mathbb{Z}/2\mathbb{Z}$ and write $\sigma \in G$ for the generator. Define an action $G \times X \rightarrow X$ by $e.v = v$ and $\sigma.v = v_0 - v$. It may be used without proof that this defines a continuous action. For which values of $v_0 \in \mathbb{R}^n$ is this action a covering space action? *Justify your answer.*

5. Let $p : X \rightarrow Y$ be a covering map.

(a) State the *path lifting theorem*.

(b) Let $y \in Y$, and let $F = p^{-1}(\{y\})$. Explain how the action of $\pi_1(Y, y)$ on F is defined.

(c) Assume X is path connected. Prove that the action of $\pi_1(Y, y)$ on $F = p^{-1}(\{y\})$ is transitive.

In this problem, it is not sufficient to state "this was proved in class" or similar.

6. For each of the following statements, mark if they are true or false.

In this problem, you do not need to justify your answers.

- (a) Any contractible space is simply connected.
- (b) For any space X and any two maps $f_0, f_1 : X \rightarrow \mathbb{R}^2$, there exists a homotopy from f_0 to f_1 .
- (c) Any subspace $A \subset \mathbb{R}^2$ is contractible.
- (d) $\mathbb{R}^n \setminus \{0\}$ is contractible for all $n \in \mathbb{N}$.
- (e) $\mathbb{R}^2 \setminus \{0\}$ is simply connected.
- (f) For any space X , the identity map $\text{id}_X : X \rightarrow X$ is a covering map.
- (g) If $p : X \rightarrow X$ is a covering map, then $p = \text{id}_X$.
- (h) There exists a covering map $p : \mathbb{R}P^2 \rightarrow S^2$.
- (i) There exists a covering map $p : \mathbb{R} \rightarrow S^1$.
- (j) If X is a discrete topological space, and $\alpha : G \times X \rightarrow X$ is a free action of a group G , then it is a covering space action.